19 $\dfrac{1+\sqrt{2}}{(3-\sqrt{2})^2}$ can be written in the form $a + b\sqrt{2}$

Find the value of a and the value of b.

Denominator: $(3 - \sqrt{2})^2$

1 mark
Expand square

×	3	$-\sqrt{2}$
3	9	$-3\sqrt{2}$
$-\sqrt{2}$	$-3\sqrt{2}$	2

$= 11 - 6\sqrt{2}$.

Multiply by the conjugate:

1 mark
Rationalise

$$\frac{11 + 6\sqrt{2}}{11 + 6\sqrt{2}}$$

Numerator: $(1 + \sqrt{2})(11 + 6\sqrt{2})$

×	11	$6\sqrt{2}$
1	11	$6\sqrt{2}$
$\sqrt{2}$	$11\sqrt{2}$	12

1 mark
Expand product

$= 23 + 17\sqrt{2}$.

Denominator: $121 - 72 = 49$.

$$\frac{23 + 17\sqrt{2}}{49}$$

$a = \dfrac{23}{49}$ $\quad$ $b = \dfrac{17}{49}$

1 mark
a=23/49

$a =$ .....

1 mark
b=17/49

$b =$ .....

(Total for Question 19 is 5 marks)